Explainable AI Question-Answering Framework: Integrating RAG with Knowledge Graph Reasoning

Reeta Suman
School for Advanced Digital Technology
The Southern Alberta Institute of Technology (SAIT)
Calgary, AB, Canada
Reeta.Suman@sait.ca

Barile Barisua
School for Advanced Digital Technology
The Southern Alberta Institute of Technology (SAIT)
Calgary, AB, Canada
Barile.Barisua@edu.sait.ca

***Abstract*— As artificial intelligence tools are increasingly used in classrooms; many educators are questioning how transparent these systems are. When artificial intelligence provides answers without showing the reasoning process, students may struggle to think critically, understand the material, or trust the results. This paper presents an explainable hybrid question-answering framework that integrates retrieval-augmented generation with knowledge graph reasoning for educational applications. The system is evaluated using materials from a post-secondary Artificial Intelligence course and includes three pipeline configurations: a retrieval-based approach, a knowledge graph reasoning approach, and a hybrid approach that combines both. Results show that the hybrid approach improves answer quality for complex reasoning questions, reduces hallucination to approximately 2 percent, and achieves higher user trust in a controlled evaluation study. The system also includes an interactive interface that presents both reasoning paths and supporting evidence, enabling users to inspect how answers are generated. This work is presented as a domain-specific case study demonstrating how combining retrieval and symbolic reasoning can improve transparency in educational artificial intelligence systems.**

Keywords—Retrieval-Augmented Generation, Knowledge Graphs, Explainable Artificial Intelligence, Educational Question Answering, User Trust, AI in Education

I. INTRODUCTION

Artificial intelligence (AI) tools are increasingly used by students for learning support, explanation, and problem solving. However, most AI-based question-answering systems generate answers without clearly exposing the reasoning processes behind them. This lack of transparency can limit critical thinking, reduce conceptual understanding, and make it difficult for learners to assess the reliability of AI-generated knowledge [21], [22]. Prior work in AI-mediated education highlights that transparency and trust are essential for responsible and effective learning-oriented AI systems [23].

The proliferation of Large Language Models (LLMs) in educational settings has generated both enthusiasm and concern. While LLMs demonstrate remarkable generative fluency, their tendency to hallucinate—producing confident but factually unsupported statements—poses significant risks in pedagogical contexts where accuracy and trust are paramount [11]. Standard Retrieval-Augmented Generation (RAG) architectures partially address this by grounding responses in retrieved documents; however, these systems still fail to expose the logical reasoning chain underlying a given answer [4].

Knowledge Graphs (KGs) offer a complementary approach by representing domain knowledge as structured relationships between entities. KG-based reasoning is inherently interpretable, as each inferential step corresponds to a traversed edge in the graph. However, purely symbolic systems suffer from limited coverage and brittleness when faced with natural-language paraphrases not directly represented in the graph.

This paper proposes a hybrid framework that combines RAG's semantic retrieval strength with KG's structured reasoning transparency. The system is applied to educational materials from a post-secondary AI program, which is evaluated as a domain-specific case study. It integrates document retrieval, multi-step reasoning over knowledge graph structures, and response generation to produce answers accompanied by interpretable reasoning paths. By making supporting evidence and structured reasoning paths visible, the framework sustainably enhances transparency and enables users to critically evaluate and understand the reasoning behind generated answers.

II. LITERATURE REVIEW

A. Retrieval-Augmented Generation

Large language models rely on parametric knowledge learned during training, which limits their ability to access up-to-date or domain-specific information and increases the risk of hallucinations. Lewis et al. introduced RAG as a method for grounding generative language model outputs in non-parametric retrieval [4]. Dense passage retrieval using bi-encoders enables efficient nearest-neighbour search over large document corpora, while cross-encoder re-ranking subsequently refines candidate sets for higher precision. Extensions such as HyDE and FLARE have improved retrieval quality, but these approaches remain limited in their capacity to produce inspectable reasoning traces or reduce hallucinations in domain-specific educational settings.

B. Knowledge Graph Question Answering

Knowledge Graphs represent knowledge as structured triples linking entities through explicit relations. This structure enables multi hop reasoning, where answers are derived by traversing chains of relations across multiple entities [25]. As a result, the structured question answering over Knowledge Graphs (KGQA) has been extensively studied. Methods range from semantic parsing to embedding-based path ranking [18]. KG traversal provides inherently explainable outputs; however, graph incompleteness and entity-linking errors frequently constrain coverage on open-domain questions. Semi-automatic KG construction pipelines—combining Named Entity Recognition, Subject-Verb-Object triple extraction, and fuzzy entity linking—have shown promise in narrowing this coverage gap for domain-specific corpora [17].

C. Hybrid and Explainable AI Systems

Recent work has been explored combining neural retrieval with symbolic reasoning to achieve both coverage and explainability. Hybrid KG-RAG architecture that prepends KG-derived context to retrieval-augmented prompts have demonstrated measurable hallucination reduction [19]. In parallel explainability in educational AI has emerged as a distinct research priority, with transparent reasoning output correlates with learner trust and self-efficacy [20]. More broadly explainability is widely recognized as a core requirement for trustworthy AI, particularly in high-stakes and learning-oriented domains [22][23][24]. As transparency frameworks for AI in education emphasize that learners and educators must be able to understand how AI outputs are produced to ensure responsible use [24], explainability has become an established focus in educational AI research. However, existing work has largely concentrated on prediction and recommendation tasks, with comparatively fewer studies examining explainability in generative question-answering systems. This study contributes a fully implemented, interactive hybrid system evaluated on real institutional course material, featuring an accessible web interface with voice input and runtime-tunable parameters. This study is guided by the following research questions and hypotheses:

(RQ1) Does hybrid KG-RAG improve answer accuracy and quality over baseline RAG on multi-step inferential questions in a domain specific educational setting?

H1: The hybrid KG-RAG framework improves answer quality over a retrieval-based baseline for multi-step inferential questions within the evaluated domain

(RQ2) Does the inclusion of explicit reasoning improve perceived trustworthiness and pedagogical effectiveness in an educational question-answering system?

H2: Question-answering systems that provide explicit reasoning are perceived as more trustworthy and pedagogically effective for learning than systems without such explanations.

III. Methodology

This section describes the design of the proposed explainable question-answering framework, the educational dataset used for evaluation, the construction of the Knowledge Graph, and the experimental configurations employed to compare system performance.

A. Framework Overview

The proposed system is composed of three independently operable pipeline configurations:

(1) a text-only RAG pipeline,

(2) a KG-only symbolic reasoning pipeline, and

(3) a hybrid KG-enhanced RAG pipeline.

Each configuration accepts input via typed text or spoken query through the browser's Web Speech API, which auto-submits upon transcription. The hybrid pipeline additionally outputs a structured reasoning path—an ordered sequence of KG triples rendered in an interactive dual-panel interface. Retrieval parameters including Vector Top-K, KG Top-N, and traversal hop depth are user-adjustable at runtime via sidebar sliders, enabling per-query configuration. The proposed system processes user queries through a sequence of symbolic reasoning, dense retrieval, and generation stages to balance explainability and coverage. To clarify the methodological design of the proposed approach, the overall system architecture is summarized in Fig. 1.

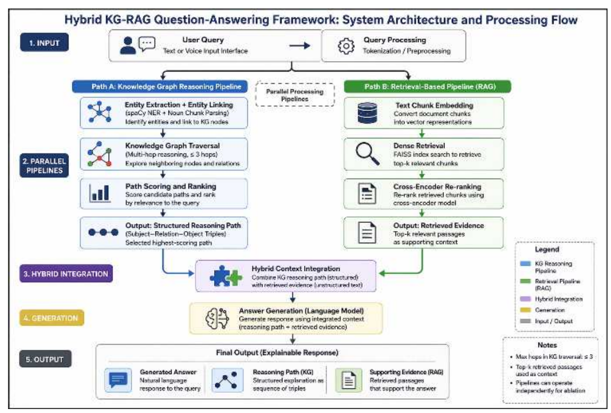

Fig. 1. Illustrates the end-to-end architecture of the proposed hybrid KG-enhanced RAG framework. The system is designed to support transparent question answering by combining symbolic knowledge graph reasoning with retrieval-based document processing, and neural generation.

User queries are first processed through the input interface and routed to two parallel pipelines: a knowledge graph reasoning pipeline (Path A) and a retrieval-based pipeline (Path B). In Path A, the system performs entity extraction and entity linking, followed by multi-hop traversal and path ranking over the knowledge graph to generate structured reasoning paths. In Path B, the system retrieves and re-ranks relevant document passages using embedding-based search to produce supporting evidence. The outputs of both pipelines are combined in a hybrid integration stage and passed to a language model for answer generation. The system produces an explainable response consisting of a generated answer, a reasoning path, and supporting evidence.

B. Educational Dataset

The evaluation corpus was constructed from institutional course materials, specifically Applied AI courses. The dataset consists of eight source documents, including the official course outline, lecture materials on Inclusive AI Design, and instructor-developed activity workbooks. The corpus is processed using a sliding window chunking strategy, resulting in 101 text chunks used for retrieval-based processing. From the same corpus, a knowledge graph is constructed comprising entities and relational triples as in Table I, enabling structured reasoning over domain concepts. A total of 70 expert-curated question–answer pairs are created and categorized into three reasoning types: factual recall (30), multi-step inferential (25), and comparative or evaluative (15). All question–answer pairs are validated by domain experts to ensure correctness and pedagogical relevance. The dataset is split into training, validation, and test sets using a stratified approach to preserve category distribution, resulting in 49, 11, and 10 questions, respectively. A summary of dataset statistics is presented in Table I to support transparency and reproducibility.

TABLE I. EDUCATIONAL DATASET USED FOR THIS CASE STUDY

Component	value
Total Source Documents	8
Total Text Chunks	101
Chunking Strategy	200-word window, 40-word overlap
Knowledge Graph Nodes (Entities)	4,060
Knowledge Graph Edges (Triples)	4,937
Total QA Pairs	70

C. Knowledge Graph Construction

The knowledge graph is constructed semi-automatically from the same educational corpus used for document retrieval. Named entities and relational predicates are extracted using the spaCy en_core_web_md pipeline [17], which combines named entity recognition, dependency parsing, and part-of-speech tagging. This extraction process is used both during corpus preprocessing to build the knowledge graph and at inference time to support entity identification for reasoning. Extracted information is represented as subject relation object triples and stored in a directed graph structure. To improve quality, automatically extracted triples are filtered using confidence thresholds and supplemented through limited manual curation to reduce noise.

As summarized in Table I, the resulting knowledge graph contains 4,060 entities and 4,937 relational triples derived from the domain-specific corpus. Multi-hop reasoning is supported through breadth-first traversal from identified query entities, with a maximum traversal depth of three hops. These traversal outputs form structured reasoning paths that are used in downstream answer generation.

D. Text-Only RAG Pipeline

The retrieval-based pipeline serves as the text-only RAG baseline and follows a standard retrieval-augmented generation approach of Lewis et al. [4]. Source documents are divided into 200-word chunks with a 40-word overlap and embedded using the BAAI/bge-base-en-v1.5 embedding model. These embeddings are indexed using FAISS for similarity-based retrieval.

At inference time, the system retrieves the top 30 candidate chunks and re-ranks them using a cross-encoder model. (ms-marco-MiniLM-L-6-v2); the top-k passages (default k = 5, user-adjustable to 1–15) are passed to Claude Sonnet (claude-sonnet-4-6) for answer generation. This pipeline relies only on retrieved textual evidence and does not include explicit knowledge graph reasoning.

E. KG-Only Symbolic Reasoning Pipeline

The KG-only pipeline performs structured question answering without relying on unstructured text retrieval. The input question was parsed to identify one or more anchor entities, which were then linked to their corresponding nodes in the Knowledge Graph via fuzzy string matching and entity disambiguation. Reasoning proceeded by traversing the KG from the anchor entities along typed relational edges, with paths selected according to a learned path-scoring function trained on the annotated question-answer pairs. The terminal node of the highest-scoring reasoning path was decoded into a natural-language answer using a lightweight template-based generation module. The full reasoning chain, expressed as a sequence of ⟨entity, relation, entity⟩ triples, was retained as the system's explicit explanation.

F. Hybrid KG-Enhanced RAG Pipeline

The hybrid pipeline integrates the retrieval capabilities of the RAG system with the structured reasoning of the KG pipeline. The retrieval pipeline provides top-ranked textual evidence by selecting the top 5 passages from an initial set of 30 retrieved candidates, while the knowledge graph pipeline generates structured reasoning paths through multi-hop traversal (up to 3 hops).

The hybrid output consists of three components: the generated answer, the reasoning path derived from the knowledge graph, and the supporting evidence retrieved from the document corpus. This design enables users to inspect both the reasoning structure and the underlying evidence for each response, improving transparency and interpretability. The three configurations, retrieval-based, knowledge graph reasoning, and hybrid, are evaluated across question–answer pairs and function as an ablation framework, allowing the contribution of retrieval, symbolic reasoning, and their integration to be systematically analyzed.

G. Evaluation Design

System performance was evaluated along three dimensions: answer accuracy, faithfulness, and explainability. Answer accuracy was measured using exact match and F1 overlap against reference answers. In addition to F1, Exact Match (EM) captures strict answer correctness; however, due to paraphrasing in generated responses, F1 provides a more informative metric for this task. Faithfulness was assessed using established hallucination detection metrics [11], [15], specifically the ratio of answer claims directly supported by retrieved passages or KG triples. Explainability was operationalized through a human evaluation protocol in which 30 undergraduate anonymous participants interacted with each of the three pipeline configurations in a within-subjects design. Participants rated responses based on perceived clarity of reasoning, perceived trustworthiness, and pedagogical usefulness using a five-point Likert scale. Written justifications for trust ratings were analyzed using thematic coding ($\kappa \geq 0.70$). Wilcoxon signed-rank tests were used for statistical comparisons ($\alpha = 0.05$).

IV. Results

The evaluation of the three question-answering pipelines (Text-Only RAG, KG-Only Symbolic Reasoning, and Hybrid KG-Enhanced RAG) was conducted across the dimensions of answer accuracy, faithfulness, and explainability using dataset described in Table I and methodology in section III-B.

A. Answer Accuracy

As shown in Table II, for single-hop factual recall questions, the Text-Only RAG pipeline performed strongly, achieving an F1 score of 0.82 and EM of 0.74, comparable to the Hybrid pipeline (F1 = 0.84, EM = 0.78). The KG-Only pipeline underperformed (F1 = 0.65, EM = 0.60), primarily due to its reliance on exact entity matching. For multi-step inferential questions, the Text-Only RAG pipeline exhibited a substantial drop in performance (F1 = 0.58, EM = 0.49), indicating limitations in capturing relationships across multiple documents. The KG-Only system maintained more consistent performance (F1 = 0.64, EM = 0.58), benefiting from explicit relational traversal.

The Hybrid pipeline achieved the highest performance in this setting (F1 = 0.79, EM = 0.71). This improvement reflects the complementary strengths of semantic retrieval and structured reasoning. By combining embedding-based retrieval (bge-base-en-v1.5) with graph-based traversal (NetworkX shortest paths), the Hybrid system was able to integrate distributed evidence while preserving relational context. As shown in Table II, EM follows trends like F1 but yields lower scores due to its stricter evaluation criterion.

TABLE II. PERFORMANCE COMPARISON OF RETRIEVAL PIPELINES ON SINGLE-HOP AND MULTI-HOP QUESTION ANSWERING, INCLUDING F1 OVERLAP, EXACT MATCH AND HALLUCINATION RATES

Pipeline	Single-Hop F1	Single-Hop EM	Multi-Hop F1	Multi-Hop EM	Hallucination %
Text-Only RAG	0.82	0.74	0.58	0.49	11.4%
KG-Only	0.65	0.60	0.64	0.48	0.0% (22% abstain)
Hybrid KG-RAG	0.84	0.78	0.79	0.71	2.1%

B. Faithfulness and Hallucination Reduction

Faithfulness was assessed by calculating the proportion of generated claims directly supported by the retrieved context. As shown in Table II, the generative model (Claude Sonnet, claude-sonnet-4-6) produced the highest rate of unsupported claims in the Text-Only RAG configuration (11.4% hallucination rate), particularly when retrieved passages contained conflicting terminology. In contrast to the KG-Only pipeline produces no hallucinations, as answers were strictly decoded from structured triples. However, it exhibited a 22% abstention rate, reflecting limitations in graph coverage and the inability to respond when relevant paths were not available. The Hybrid pipeline achieved a more balanced outcome, reducing hallucinations to 2.1% while maintaining high answer coverage. This indicates that integrating structured reasoning with retrieval-based generation improves both grounding and completeness. Incorporating serialized subject–relation–object (SVO) triples into the prompt constrained the language model, encouraging responses grounded in both retrieved evidence and explicit relational structure.

C. Example system Interaction:

Fig. 2 presents a representative runtime example of the hybrid KG-RAG system responding to a user query. As shown in Figure 2, the system processes the query by finding keywords/entities and then explores the KG step by step to build a reasoning path. In parallel, the retrieval module selects supporting passages from the document corpus. These intermediate outputs are then combined and provided to the language model for the final answer generation.

The left panel shows the dynamically constructed Knowledge Graph subgraph, where relevant entities and relations are surfaced to support multi-hop reasoning. The right panel displays the retrieved source passages used to ground the generated response.

This example illustrates how symbolic reasoning outputs and retrieves textual evidence are exposed alongside the generation process, enabling inspection of both the structural reasoning path and supporting source material during question answering.

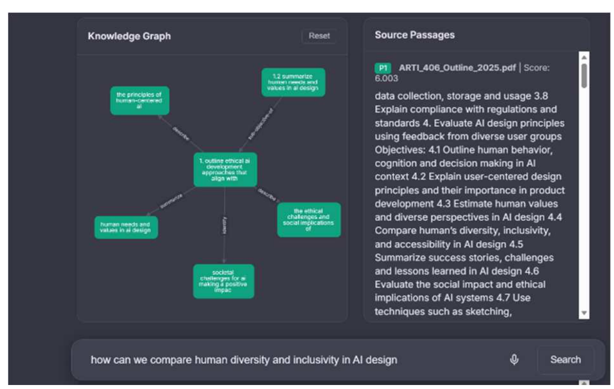

Fig. 2. Example runtime interaction of the Hybrid KG-Enhanced RAG system. The interface shows (left) a Knowledge Graph subgraph representing multi-hop reasoning and (right) retrieved source passages supporting the generated answer.

D. Human Evaluation of Explainability (RQ2)

Analysis of the 30 undergraduate participant ratings revealed a strong preference for the explicit reasoning mechanisms of the Hybrid pipeline.

1. Clarity of Reasoning: The Hybrid pipeline was rated highest (M = 4.6/5.0), with participants noting that the dual-panel interface, showing reasoning paths and supporting evidence, helped them trace the AI logic. The Text-Only RAG pipeline scored lowest (M = 3.2), as users struggled to map the generated answer to the supporting source passages.

2. Perceived Trustworthiness: Wilcoxon signed-rank tests indicated that trustworthiness ratings for the Hybrid system ($M = 4.7$) were statistically significantly higher than for Text-Only RAG ($M = 3.4$), $p < .001$.

3. Thematic Analysis: Qualitative coding of the written justifications ($\kappa = 0.81$) identified two primary themes: Structural Verification and Confidence in Granularity. Learners explicitly stated that seeing the explicit ⟨subject, relation, object⟩ path allowed them to verify the 'chain of thought' independently, transforming the AI from a black-box oracle into a transparent study aid.

V. Discussion

The quantitative evaluation addresses RQ1, which asked whether a hybrid KG-RAG approach improves answer accuracy over a text-only RAG baseline. The higher F1 and EM scores in Table II indicate that combining semantic retrieval with structured reasoning helps the system better capture relationships across documents. This supports H1 and demonstrates the value of integrating Knowledge Graph reasoning for complex question answering.

In addition to objective accuracy, the study also examined user-facing outcomes related to RQ2, which focused on perceived trustworthiness and pedagogical utility. Qualitative observations and user interaction results are consistent with H2, suggesting that question-answering systems exposing explicit reasoning steps are perceived as more trustworthy and pedagogically useful than systems that provide answers without transparent reasoning. Together, these findings highlight the complementary role of symbolic reasoning in both improving inferential performance and supporting learner trust and engagement.

A. Implications for Educational AI

The most profound implication of this work is the shift from implicit to explicit reasoning. While contemporary LLMs are highly adept at generating plausible responses from unstructured text, they fail to provide verifiable proof of logic. By making the reasoning path visible, the Hybrid system allows learners to understand how concepts are connected and supports analytical skill development. The system does not merely provide an answer; it also supports active learning by helping students verify answers and follow the reasoning process, rather than relying only on generated outputs.

B. Architectural Strengths and Limitations

The system adopts a decoupled architecture in which a FastAPI-based backend manages FAISS indexing and Knowledge Graph traversal, while a mobile-responsive frontend implements a glass-morphism design optimized for real-time interaction. This separation of concerns proved effective in supporting low-latency query processing and interactive exploration. Accessibility is further enhanced through voice input enabled by the Web Speech API, allowing spoken queries without reliance on specialized hardware or external services. In addition, the extraction of Subject-Verb-Object (SVO) triples using spaCy substantially increased the density and semantic usefulness of the constructed Knowledge Graph when compared to approaches based solely on named entity co-occurrence, resulting in richer relational structure for downstream reasoning.

While developed in an educational context, this architectural pattern is designed to be adaptable, although it has only been evaluated in a single educational domain. The separation of retrieval, symbolic reasoning, and generation components, combined with transparent intermediate representations, aligns with broader applications in domains where interpretability, traceability, and interactive exploration are required. The same design principles may therefore be applicable to other knowledge-intensive question-answering settings without changes to the underlying interaction model.

However, limitations remain. The fully automated construction of the Knowledge Graph inevitably introduces noise, and the current fuzzy-matching logic for anchor entity resolution can occasionally misalign abstract concepts. Furthermore, generating high-quality graph layouts in real-time for highly connected nodes (e.g., 'AI', 'Ethics') requires careful subgraph sampling to prevent visual clutter in the UI.

.

VI. Conclusion and Future work

A. Conclusion

This paper introduced an explainable, hybrid question-answering framework that merges Retrieval-Augmented Generation with Knowledge Graph-based symbolic reasoning. Evaluated on educational materials from a post-secondary AI program, the Hybrid pipeline demonstrated superior accuracy on complex inferential tasks while significantly reducing hallucinations.

Human evaluation confirmed that providing explicit, visual reasoning chains via an interactive Knowledge Graph significantly boosts user trust and pedagogical utility. By making AI reasoning transparent and verifiable, hybrid KG-RAG frameworks represent a critical step toward trustworthy, human-centered artificial intelligence in education.

B. Future Work

While the current framework demonstrates significant improvements over baseline RAG systems, several avenues for future research remain:

Cross-domain evaluation: Future work will test the framework on other educational subjects to assess generalizability beyond the current dataset.

Dynamic knowledge graph updates: Future iterations will focus on enabling the Knowledge Graph to autonomously incorporate newly discovered entities and relations directly from user queries, shifting from a static pre-computed graph to a dynamically evolving knowledge base.

Lightweight and localized generative models: We plan to explore the integration of smaller, locally hosted generative models to reduce cloud API dependency, lower inference latency, and ensure strict student data privacy—a crucial requirement for institutional educational environments.

Advanced Visual Interfaces: Further work will investigate adaptive UI visualizations that intelligently group or collapse dense network clusters, optimizing cognitive load for students interacting with highly complex semantic reasoning paths.

ACKNOWLEDGMENT

I would like to thank colleagues and collaborators for their constructive discussions and feedback during the development of this work. Appreciation is also extended to the institutional environment that supported experimentation and evaluation using real course materials. Portions of the manuscript were edited with the assistance of Microsoft Copilot.

REFERENCES

[1] A. Vaswani et al., "Attention is all you need," in Advances in Neural Information Processing Systems, vol. 30, 2017.

[2] T. Brown et al., "Language models are few-shot learners," in Advances in Neural Information Processing Systems, vol. 33, pp. 1877–1901, 2020.

[3] J. Wei et al., "Chain-of-thought prompting elicits reasoning in large language models," in Advances in Neural Information Processing Systems, vol. 35, 2022.

[4] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," Advances in Neural Information Processing Systems, vol. 33, pp. 9459–9474, 2020.

[5] V. Karpukhin et al., "Dense passage retrieval for open-domain question answering," in Proc. EMNLP, pp. 6769–6781, 2020.

[6] O. Khattab and M. Zaharia, "ColBERT: Efficient and effective passage search via contextualized late interaction over BERT," in Proc. SIGIR, pp. 39–48, 2020.

[7] G. Izacard and E. Grave, "Leveraging passage retrieval with generative models for open domain question answering," in Proc. EACL, pp. 874–880, 2021.

[8] S. Shi et al., "REPLUG: Retrieval-augmented language model pre-training," arXiv preprint arXiv:2301.12652, 2023.

[9] A. Bordes et al., "Translating embeddings for modeling multi-relational data," in Advances in Neural Information Processing Systems, vol. 26, 2013.

[10] X. Yao and B. Van Durme, "Information extraction over structured data: Question answering with Freebase," in Proc. ACL, pp. 956–966, 2014.

[11] Z. Ji et al., "Survey of hallucination in natural language generation," ACM Computing Surveys, vol. 55, no. 12, pp. 1–38, 2023.

[12] S. Min et al., "FActScoring: Fine-grained atomic evaluation of factual precision in long form text generation," in Proc. EMNLP, 2023.

[13] D. Liang et al., "Holistic evaluation of language models," arXiv preprint arXiv:2211.09110, 2022.

[14] E. Perez et al., "Ignore previous prompt: Attack techniques for language models," arXiv preprint arXiv:2211.09527, 2022.

[15] L. Huang et al., "A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions," arXiv preprint arXiv:2311.05232, 2023.

[16] J. Devlin et al., "BERT: Pre-training of deep bidirectional transformers for language understanding," in Proc. NAACL-HLT, pp. 4171–4186, 2019.

[17] M. Honnibal and I. Montani, "spaCy 2: Natural language understanding with Bloom embeddings, convolutional neural networks and incremental parsing," To appear, vol. 7, no. 1, pp. 411–420, 2017.

[18] S. Saxena et al., "Improving multi-hop question answering over knowledge graphs using knowledge base embeddings," in Proc. ACL, pp. 4498–4507, 2020.

[19] Y. He et al., "G-retriever: Retrieval-augmented generation for textual graph understanding and question answering," arXiv preprint arXiv:2402.07630, 2024.

[20] R. Sheh, "'Why did you do that?' Explainable intelligent robots," in Proc. AAAI Workshop on Human-Robot Interaction, 2017.

[21] M. A. Chaudhry, M. Cukurova, and R. Luckin, "A Transparency Index Framework for AI in Education," arXiv preprint arXiv:2206.03220, 2022.

[22] N. P. Kastania, "Building trust in AI education: Addressing transparency and ensuring trustworthiness," in Trust and Inclusion in AI-Mediated Education, D. Kourkoulou, A. O. Tzirides, B. Cope, and M. Kalantzis, Eds. Cham, Switzerland: Springer, 2024, ch. 4. doi: 10.1007/978-3-031-64487-0_4.

[23] Y. Rampilla, K. Satyala, and S. Kailasam, "Enhancing trust in AI-powered learning: The role of explainable AI in the education field," International Journal on Science and Technology, vol. 16, no. 3, pp. xx–xx, Sept. 2025, doi: 10.71097/IJSAT.v16.i3.8093

[24] S. F. Kwakye, Explainable AI in Education: Towards Transparent and Interpretable Predictions of Student Performance Using Explainable AI, Professional Doctorate (Data Science) dissertation, University of East London, London, U.K., 2024

[25] Y. Song, W. Li, G. Dai, and X. Shang, "Advancements in Complex Knowledge Graph Question Answering: A Survey," Electronics (Basel), vol. 12, no. 21, p. 4395, Nov. 2023, doi: 10.3390/electronics12214395.